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AI25BTECH11010 - Dhanush Kumar

Question : The eigenvalues of a 2×2 matrix $\mathbf{X}$ are -2 and -3 . Find the eigenvalues of the matrix $(\mathbf{X} + \mathbf{I})^{-1}(\mathbf{X} + 5\mathbf{I})$.

(a) $-3, -4$

(c) $-1, -3$

(b) $-1, -2$

(d) $-2, -4$

Solution:

$$\text{Let } \mathbf{X}\mathbf{v} = \lambda\mathbf{v}, \text{ where } \mathbf{v} \text{ is an eigenvector of } \mathbf{X}. \quad (1)$$

$$(\mathbf{X} + \mathbf{I})^{-1}(\mathbf{X} + 5\mathbf{I})\mathbf{v} = (\mathbf{X} + \mathbf{I})^{-1}[(\mathbf{X} + \mathbf{I}) + 4\mathbf{I}]\mathbf{v} \quad (2)$$

$$= (\mathbf{X} + \mathbf{I})^{-1}(\mathbf{X} + \mathbf{I})\mathbf{v} + 4(\mathbf{X} + \mathbf{I})^{-1}\mathbf{v} \quad (3)$$

$$= \mathbf{v} + 4(\mathbf{X} + \mathbf{I})^{-1}\mathbf{v}. \quad (4)$$

Since

$$(\mathbf{X} + \mathbf{I})\mathbf{v} = (\lambda + 1)\mathbf{v}, \quad (5)$$

$$\Rightarrow (\mathbf{X} + \mathbf{I})^{-1}\mathbf{v} = \frac{1}{\lambda + 1}\mathbf{v}. \quad (6)$$

Substitute this into the earlier expression:

$$(\mathbf{X} + \mathbf{I})^{-1}(\mathbf{X} + 5\mathbf{I})\mathbf{v} = \left(1 + \frac{4}{\lambda + 1}\right)\mathbf{v} \quad (7)$$

$$= \frac{\lambda + 5}{\lambda + 1}\mathbf{v}. \quad (8)$$

Hence, each eigenvalue λ of $\mathbf{X}$ maps to

$$\mu = \frac{\lambda + 5}{\lambda + 1}. \quad (9)$$

For the given eigenvalues of $\mathbf{X}$:

$$\lambda_1 = -2, \quad \lambda_2 = -3, \quad (10)$$

we get

$$\mu_1 = \frac{-2 + 5}{-2 + 1} = \frac{3}{-1} = -3, \quad (11)$$

$$\mu_2 = \frac{-3 + 5}{-3 + 1} = \frac{2}{-2} = -1. \quad (12)$$

Eigenvalues of $(X + I)^{-1}(X + 5I)$ are -3 and -1 .

Answer: (c) $-1, -3$